EEEN20020 Experiment 2 — Bench data sheet

Names: _____ / _____ Bench: _____ Date: ____ / ____ / 2026

Design: R1max = 50 kΩ (trimmer code 503) · R2 = 10 kΩ · R3 = 10 kΩ · expected A_{Vmin} = 1.00, A_{Vmax} = 6.00 (**must measure ≥ 5**)

1. Components — measure OUT OF CIRCUIT before inserting (needed for Section G)

Item	Nominal	Measured (DMM)	Notes
R1max — pot at max (wiper ↔ end used)	50 kΩ		
R1min — pot at min (wiper ↔ end used)	≈ 0 Ω		
Pot end-to-end	50 kΩ		
R2	10 kΩ		bands:
R3	10 kΩ		bands:

2. Supplies and set-up

Pin 7 (V_{S+})	Pin 4 (V_{S-})	Pin 6 DC, no input	PSU current (+ / −)	Generator: 1 kHz · 1.000 Vpp · 0 V offset · HighZ
			/	☐ checked

Table I colours: V_{S+} _____ V_{S-} _____ GND _____ v_{in} _____ v_{out} _____

3. Measurements (gain = CH2 ÷ CH1, from the SAME screen)

Test	V/div CH1 / CH2	t/div	Vpp CH1 (in)	Vpp CH2 (out)	Gain	RMS CH1	RMS CH2	RMS gain	Photo
Sine, min gain	/								P3
Sine, max gain	/								P4
Channel-swap check (min)	/								—
Bench, min gain (STOP)	/								P5
Bench, max gain (STOP)	/								P6

Headphones connected during bench captures? ☐ Yes ☐ No

4. Audio and Demonstrator

Is the signal audible and free of distortion? (Demonstrator's answer)	☐ Yes ☐ No	Initials:
What we heard at A_V = 1		
What we heard at A_{Vmax}		

5. Photo shot-list (two of each, check focus before dismantling)

☐ P1 breadboard top-down ☐ P2 whole set-up (breadboard + PSU + generator + scope) ☐ P3 sine min ☐ P4 sine max ☐ P5 bench min ☐ P6 bench max ☐ P7 DMM readings (optional)

6. Before leaving the bench

☐ $A_{Vmin} \approx 1.00$ ☐ $\mathbf{A_{Vmax} \geq 5}$ ☐ peaks round (no clipping) ☐ every number has a photo ☐ R's measured ☐ Demonstrator ticked ☐ equipment returned ☐ upload within 1 h — **both** students